

Name: _____

6.1 Perpendicular and angle bisectors.

1. P lies on the perpendicular bisector of $\overline{AB}$. $PA = 5x + 1$ and $PB = 8x - 14$. Find x and PA .
2. $\overline{CM}$ is the perpendicular bisector of $\overline{AB}$, $CA = 15$, and $AM = 9$. Find CM .
3. $\overline{BD}$ bisects $\angle ABC$, $m\angle ABD = (7x + 2)^\circ$, and $m\angle DBC = (9x - 12)^\circ$. Find x and $m\angle ABC$.
4. Write an equation of the perpendicular bisector of the segment with endpoints $A(-1, 6)$ and $B(7, 2)$.
5. Point T is 11 in. from U and 11 in. from V. What can you conclude, and by which theorem?

6.2 Bisectors of triangles.

6. Find the circumcenter of $\triangle ABC$ with $A(0, 0)$, $B(0, 14)$, and $C(48, 0)$, and its distance to each vertex.
7. Q is the circumcenter of $\triangle RST$, $QR = 4x + 5$, and $QS = 7x - 13$. Find x and QT .
8. P is the incenter of $\triangle DEF$. Its distance to $\overline{DE}$ is $2y + 19$ and its distance to $\overline{EF}$ is $6y - 5$. Find y and the common distance.
9. Where does the circumcenter lie for an acute, a right, and an obtuse triangle?
10. Which center would you use to place a statue the same distance from the three straight walls of a triangular room? Explain.

6.3 Medians and altitudes.

11. P is the centroid of $\triangle ABC$ and $\overline{AX}$ is a median with $AP = 24$. Find PX and AX .
12. $\overline{BY}$ is a median with $BP = 6x$ and $PY = x + 16$, where P is the centroid. Find x and BY .
13. Find the centroid of $\triangle DEF$ with $D(-6, 1)$, $E(2, 9)$, and $F(7, -4)$.
14. Where is the orthocenter of a right triangle? Of an obtuse triangle?

6.4 The Triangle Midsegment Theorem.

15. $\overline{DE}$ is a midsegment of $\triangle ABC$ parallel to $\overline{AB}$. If $AB = 62$, find DE . If $DE = 7x - 4$ and $AB = 10x + 10$, find x .
16. $\triangle PQR$ has $P(-6, 2)$, $Q(0, 8)$, and $R(4, -2)$. Find the midpoints M of $\overline{PQ}$ and N of $\overline{PR}$, then show $\overline{MN} \parallel \overline{QR}$.
17. The perimeter of $\triangle ABC$ is 72. Find the perimeter of its midsegment triangle.

6.5 Indirect proof and inequalities.

18. Order the angles of $\triangle ABC$ from smallest to largest given $AB = 17$, $BC = 10$, and $CA = 13$.
19. Can 8, 12, and 21 be the sides of a triangle? Can 6, 9, and 13? Show the check.
20. Two sides of a triangle measure 11 and 26. Find the range for the third side. Then write the first sentence of an indirect proof of "a triangle cannot have two right angles."